Work Specifications and Market Rates

Lot Title: PROJECT: PROVIDING, INSTALLATION, TESTING & COMMISSIONING OF 50 KW HYBRID SOLAR SYSTEM WITH GREEN METER AT PIDCL HEAD OFFICE ISLAMABAD

Specifications / Requirements:

Sr. No	S.No.	Description	Unit	Qty	Rate	Amount
1	1	SOLAR PANELS: Providing, installation testing & commissioning of solar panels Tier -1 A grade N-type bi- Facial 720 Watts solar Panels. Make: LONGI/CANADIAN /JINKO/INVEREX	Nos.	70		
2	2	INVERTER: Supply, installation, testing, and commissioning of 16 kW Industrial Three-Phase Hybrid Solar Inverter, IP65, Make: INVEREX, GOODWE, SOLIS or equivalent	Nos.	3		
3	3	BATTERIES: Supply, installation, testing, and commissioning of High-Capacity 16 KWh (51.2V, 280Ah/ 300Ah) Lithium-Ion (LiFePO4) Battery Modules (6000-8000 cycles) Including BMS. Make: /INVEREX/Goodwe/ Pylontech or equivalent IP-65	Nos.	3		
4	4	STRUCTURE TYPE: Supply and installation of H-Beam 12-Gauge Girder & 16-Gauge Galvanized MS Mounting Frame coated with epoxy paint & red oxide primer, fixed in 1:2:4 concrete foundation pads. Complete with SS hardware and aluminum clamps.	Nos.	70		

Sr. No	S.No.	Description	Unit	Qty	Rate	Amount
5	5	DC/AC D.B & FUSES: Supply, installation, testing, and commissioning of heavy-duty MS DC Distribution Box equipped with 1000V DC Fuses, 4-Pole 125A AC main breaker & 4-POLE 125 Amps manual changeover Make: Schneider / CNC / ABB or equivalent.	Job	1		
6	6	SPD-AC: Supply, installation, testing, and commissioning of Type-2 AC Surge Protection Device (4-Pole, 40kA, 280-415V in MS enclosure. Make: CNC / Chint or equivalent	Nos.	2		
7	7	SPD-DC: Supply, installation, testing, and commissioning of Type-2 DC Surge Protection Device (2-Pole, 40kA, 1000V DC) per string. Make: CNC / Chint or equivalent	Nos.	1		
8	8	DC CABLES FOR PANEL: 6 sq mm ,1 core, flexible Dual coated DC cable. Make; Pakistan/ Fast Cables/Newage	Mtr	200		
9	9	AC CABLES FOR INVERTER: 35sq mm, -4 core PVC/ XLPE 600/1000V /CU: cable. Make Pakistan/ Fast Cables/Newage (ITEM no 82 CSR 2024)	Mtr	100		
10	10	AC CABLES FOR LOAD DISTRIBUTION: 6 SQ MM, 1 core PVC/ XLPE 600/1000V /CU: cable. Make Pakistan/ Fast Cables/Newage	Mtr	150		
11	11	EARTHING CABLES FOR SOLAR PANELS: 16 sq mm, 1-Core PVC green/yellow insulated grounding cable. Make: Pakistan/ Fast Cable/Newages	Mtr	50		

Sr. No	S.No.	Description	Unit	Qty	Rate	Amount
12	12	EARTHING CABLES FOR INVERTER & LIGHTNING ARRESTER: 16 sq mm, 1-Core PVC green/yellow insulated grounding cable. Make: Pakistan/ Fast Cable/Newages	Mtr	50		
13	13	Providing & Installation channel patti .1” as required as per instruction of EI. (schedule Rate GOS S.No 250)	Ft.	40		
14	14	Supply, installation, testing, and commissioning of Early Streamer Emission (ESE) Lightning Arrester complete with 4m GI elevation mast, or as per site requirement with guy- wire support, 16 sq mm copper down conductor, IP67 lightning strike counter, and dedicated chemical earthing pit (R 5 Ω)	Job	1		
15	15	EARTHINGS: Providing & Fixing Earthing set with 2’X2’X1/8” Copper plate buried in the Ground at a depth of 25 feet or less if water comes out from the ground level (with salt & charcoal, or earthing chemical powder) etc making the pit 25 feet deep by excavation of all type soil, fixing of 8 SWG copper wire in 1/2” PVC conduit complete in all respect. Less than 5 ohms	Nos.	3		
16	16	Net Metering Submission of File Charges (Without IESCO Challan/Estimate)	Job	1		
17		TOTAL AMOUNT (PKR)				12627950
					GRAND TOTAL:	Rs12,627,950.00

BILL OF QUANTITIES (BOQ)

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5	DC/AC D.B & FUSES Supply, installation, testing, and commissioning of heavy-duty MS DC Distribution Box equipped with 1000V DC Fuses, 4-Pole 125A AC main breaker & 4-POLE 125 Amps manual changeover Make: Schneider / CNC / ABB or equivalent.	JOB	1		
6	SPD-AC Supply, installation, testing, and commissioning of Type-2 AC Surge Protection Device (4-Pole, 40kA, 280-415V in MS enclosure. Make: CNC / Chint or equivalent	Nos.	2		
7	SPD-DC Supply, installation, testing, and commissioning of Type-2 DC Surge Protection Device (2-Pole, 40kA, 1000V DC) per string. Make: CNC / Chint or equivalent	Nos.	1		
8	DC CABLES FOR PANEL 6 sq mm ,1 core, flexible Dual coated DC cable Make: Pakistan/ Fast Cables/Newage	Mtr	200		
9	AC CABLES FOR INVERTER 35sq mm, -4 core PVC/ XLPE 600/1000V /CU: cable Make Pakistan/ Fast Cables/Newage (ITEM no 82 CSR 2024)	Mtr	100		
10	AC CABLES FOR LOAD DISTRIBUTION 6 SQ MM, 1 core PVC/ XLPE 600/1000V /CU: cable Make Pakistan/ Fast Cables/Newage	Mtr	150		
11	EARTHING CABLES FOR SOLAR PANELS 16 sq mm, 1-Core PVC green/yellow insulated grounding cable. Make: Pakistan/ Fast Cable/Newages	Mtr	50		
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13	Providing & Installation channel patti .1" as required as per instruction of EI. (schedule Rate GOS S.No 250)	Ft.	40		
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				TOTAL AMOUNT (PKR)	